

# Technology Stack

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 15 March 2026                                           |
| Team ID       | UNM293SmartStudyAI                                      |
| Project Name  | ComicCraft - AI Comic Story Creator using Gemini Models |
| Maximum Marks | 2 Marks                                                 |

**Define Your Technology Stack:**

The technology stack includes the programming languages, frameworks, databases, development tools, hosting services, and APIs used to build ComicCraft. These technologies support the creation of comic stories with the help of Google Gemini models.

The following table lists the selected technologies for each part of the project and explains their purpose.

**Technology Stack Details**

| S.No | Architecture Component / Layer                    | Technology Chosen     | Justification / Purpose                                                                                              |
|------|---------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Frontend / Client-Side</b><br>User interface   | HTML, CSS, JavaScript | Creates a simple, responsive interface where users can enter prompts, view generated stories, and read comic panels. |
| 2    | <b>Backend / Server-Side</b><br>Application logic | Python with Flask     | Handles user requests, connects the interface to Gemini, and organizes the generated story content.                  |
| 3    | <b>Database / Data Storage</b>                    | Firebase Firestore    | Stores user projects, prompts, and generated comic-story details in a flexible cloud database.                       |
| 4    | <b>Cloud / Hosting / Deployment</b>               | Firebase Hosting      | Makes the web application accessible online and supports straightforward deployment.                                 |
| 5    | <b>Version Control &amp; CI/CD</b>                | Git and GitHub        | Tracks code changes, supports team collaboration, and keeps the project source organized.                            |
| 6    | <b>Third-Party APIs / Other Tools</b>             | Google Gemini API     | Generates comic plots, dialogue, character descriptions, and scene ideas from user prompts using Gemini models.      |